

```
pip install numpy
```

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)

data visulaization is a technique used to reperesnt the data in the form of graph

```
import numpy as np
```

```
#2d-array
a=np.array([[1,2,3,4],
            [5,6,7,8],
            [9,1,3,4]
            ])
print(a)
```

```
[[1 2 3 4]
 [5 6 7 8]
 [9 1 3 4]]
```

```
#3d-array
c=np.array([
    [
        [1,2,3],
        [4,3,2]
    ],
    [
        [4,5,6],
        [5,4,2]
    ]
])
print(c)
```

```
[[[1 2 3]
   [4 3 2]]

  [[4 5 6]
   [5 4 2]]]
```

```
#to print ones
a=np.ones([2,3])
print(a)
```

```
[[1. 1. 1.]
 [1. 1. 1.]]
```

```
#to print zeroes
d=np.zeros((2,3))
print(d)
```

```
[[0. 0. 0.]
 [0. 0. 0.]]
```

```
#to print number
d=np.full([3,3],5)
print(d)
```

```
[[5 5 5]
 [5 5 5]
 [5 5 5]]
```

```
#eye
e=np.eye(4)
print(e)
```

```
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
```

```
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
```

```
#arrange
f=np.arange(2,10,3)
print(f)
```

```
[2 5 8]
```

```
#linspace
np.linspace(2,10,4)
```

```
array([ 2.          ,  4.66666667,  7.33333333, 10.          ])
```

```
from IPython.lib.security import random
if (random<=0.33):
    print("rock win")
elif(random<=0.66):
    print("paper win")
else:
    print("scissor win")
```

```
#uniform random
np.random.randint(1,5,size=(3,3))
```

```
array([[4, 1, 3],
       [1, 4, 3],
       [2, 3, 3]])
```

```
np.empty((3,5))
```

```
array([[0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0.]])
```

```
arr=np.array([["a","b","c","d","e"]])
print(arr.shape)
print(arr.size)
print(arr.ndim)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
(1, 5)
5
2
<U1
4
20
```

```
#1d-indexing
a=np.array([10,20,30,40,50])
print(a[0])
print(a[2])
print(a[-1])
```

```
10
30
50
```

```
#slicing
print(a[1:4])
print(a[:3])
print(a[::2])
print(a[::-1])
```

```
[20 30 40]
[10 20 30]
```

```
[10 30 50]
[50 40 30 20 10]
```

```
a=np.array([[1,2,3,4,5],[6,7,8,9,10]])
print(a[1,1:4])
```

```
[7 8 9]
```

```
#masking
a=np.array([10,25,8,42,17])
mask=a>15
print(mask)
print(a[mask])
print(a[a>15])
print(a[[0,3]])
```

```
[False  True False  True  True]
[25 42 17]
[25 42 17]
[10 42]
```

```
for i in range(3):
    print(i)
```

```
0
1
2
```

```
# boolean mask
a = np.array([5, 15, 25, 35, 45])

mask = a >= 25
print(mask)
print(a[mask])

# short hand
print(a[a >= 25])

# fancy indexing
```

```
[False False  True  True  True]
[25 35 45]
[25 35 45]
```

```
# Array arithmetic operations
import numpy as np

x = np.array([2, 4, 6])
y = np.array([1, 3, 5])

print(x + y)
print(x - y)

print(x / y)

print(x ** 2)
print(x + 20)
```

```
[ 3  7 11]
[1 1 1]
[2.         1.33333333 1.2       ]
[ 4 16 36]
[22 24 26]
```

```
# problem-2
a = np.eye(4).reshape(4,4)
print(a)
```

```
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
```

```
[0. 0. 1. 0.]  
[0. 0. 0. 1.]
```

```
# Problem-3  
nums = np.array([12, 5, 18, 7, 3, 15, 10, 6])  
  
print(np.max(nums))
```

```
18
```

```
print(np.argmax(nums))
```

```
2
```

```
# problem-5  
m = np.array([[2,7,1],[9,4,6],[5,8,3]])  
  
mask = m > 5  
print(m[mask])
```

```
[7 9 6 8]
```